Georgios Syros

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Education

Northeastern University, Boston, MA, United States 2023 - Present
Ph.D. in Computer Science, **GPA: 4.0**
Advisors: *Prof. Alina Oprea*, *Prof. Cristina Nita-Rotaru*

Northeastern University, Boston, MA, United States 2023 - 2025
M.Sc. in Computer Science, **GPA: 4.0**

Athens University of Economics and Business, Athens, Greece 2019 - 2023
B.Sc. in Computer Science, **GPA: 9.35/10**

Research Experience

Cybersecurity & Privacy Institute – Northeastern University, Boston, MA 2023 - Present
Doctoral Researcher
- Studying the security and robustness of AI-driven systems.
- Designing automated red-teaming frameworks for Agentic AI systems.
- Designing effective, system-level defenses for Agentic AI systems that enable user oversight and control.

Microsoft Research & AI Red Team – Microsoft Corporation, Redmond, WA Jun-Aug 2026
Research Intern
- Investigated backdoor attacks that train reasoning-LLM coding agents to issue malicious tool calls.
- Designed the attack methodology and engineered the custom fine-tuning and Reinforcement Learning pipeline.
- Evaluated attacks across multiple models, agent harnesses, and persistent malware payloads.

Maryland Cybersecurity Center – University of Maryland, College Park, MD Aug-Oct 2022
Undergraduate Researcher - Intern
- Analyzed failure modes of Machine Learning components in Database Management Systems.
- Designed & conducted performance degradation attacks on Reinforcement Learning-based Query Optimizers.

Mobile Multimedia Lab – Athens University of Economics and Business, Greece 2021 - 2023
Undergraduate Researcher
- Designed a decentralized system on Ethereum & IPFS evolvable assets and trustless royalty payouts.

Publications

Kerkoporta: Injecting Structured-Output Backdoors in Reasoning Coding Agents, preprint, 2026
Georgios Syros, Blake Bullwinkel, Alina Oprea, Cristina Nita-Rotaru, Amanda Minnich, Xiaodong Liu, Giorgio Severi

MUZZLE: Adaptive Agentic Red-Teaming of Web Agents Against Indirect Prompt Injection Attacks, USENIX Security, 2026
Georgios Syros, Evan Rose, Brian Grinstead, Christoph Kerschbaumer, William Robertson, Cristina Nita-Rotaru, Alina Oprea

SAGA: A Security Architecture for Governing AI Agentic Systems, NDSS, 2026
Georgios Syros, Anshuman Suri, Jacob Ginesin, Cristina Nita-Rotaru, Alina Oprea

DROP: Poison Dilution via Knowledge Distillation for Federated Learning, preprint, 2025
Georgios Syros*, Anshuman Suri*, Farinaz Koushanfar, Cristina Nita-Rotaru, Alina Oprea

Backdoor Attacks in Peer-to-Peer Federated Learning, ACM Transactions on Privacy and Security (TOPS), 2024
Georgios Syros*, Gokberk Yar*, Simona Boboila, Cristina Nita-Rotaru, Alina Oprea

Decentralized NFT-based Evolvable Games, 4th Conference on Blockchain Research and Applications for Innovative Networks and Services (BRAINS), 2022
Christos Karapapas, **Georgios Syros**, Iakovos Pittaras, George C. Polyzos

Talks

"MUZZLE: Adaptive Agentic Red-Teaming of Web Agents Against Indirect Prompt Injection Attacks"
@ *35th USENIX Security Symposium, Baltimore* Aug 2026
@ *Khoury Security and Privacy Day, Northeastern University, Boston* Apr 2026
"Toward Automated Holistic Red-Teaming of Web Agents"
@ *New England Security Day, UMass, Amherst* Mar 2026
"SAGA: A Security Architecture for Governing AI Agentic Systems"
@ *Network and Distributed System Security (NDSS) Symposium 2026, San Diego, CA* Feb 2026
"From Autonomy to Accountability: Securing Agentic AI Systems with SAGA"
@ *Khoury Security Day '25*, May 2025
"DROP: Poison Dilution via Knowledge Distillation for Federated Learning"
@ *New England Systems Day '25*, Feb 2025

Teaching Experience

Northeastern University
Assisted students with their projects and graded assignments for the following courses:
· CS3650 Computer Systems Fall 2025
· CS6620 Fundamentals of Cloud Computing Summer 2025

Awards and Honors

· Student Grant Award, *USENIX Association*, 2026
· Financial Scholarship for Academic Excellence, *Huawei Hellas Enterprise*
· Honor for High Academic Performance, *Athens University of Economics and Business*
· Selected for Greek delegation, *Huawei Seeds for the Future 2021*

Academic Service

Served on the Program Committee:
· ACM Workshop on Artificial Intelligence and Security (AI Sec'26)
· ACM Conference on AI and Agentic Systems (CAIS'26)

Reviewed submissions for:
· USENIX Security Symposium (USENIX SEC)
· ACM Conference on Computer and Communications Security (CCS)
· ACM Transactions on Privacy and Security (TOPS) Journal

* Equal contribution.